

Prediction and Pixel-Space Control from Ridge Regression in Linear Feature Spaces

Theory note, revised with deterministic-equivalent feature-space analysis

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Abstract

We study a noisy linear teacher and a ridge-regression student. The student may be trained after a linear feature map, but accentuation is performed by back-propagating its gradient to the input pixels. The feature map creates two distinct geometries: the feature covariance $C = A\Sigma A^\top$ controls natural-image prediction, whereas $G = AA^\top$ controls pixel-space gradient motion. This distinction gives closed-form finite-sample expressions for generalization, own-path accentuation, and peer review. In a spectral feature map aligned with the eigenvectors of Σ , random-matrix deterministic equivalents (DEs) give per-mode weight bias and variance, generalization error, accentuation error, slopes, and both R^2 measures. The same DE risk can also approximate cross-validation and select the ridge penalty. Full PCA, whitening, and top-PC truncation emerge as transparent special cases.

1 Teacher, features, and ridge student

Let the centered input and noisy response be

$$\mathbf{x} \sim \mathcal{N}(0, \Sigma), \quad y = (\boldsymbol{\beta}^\star)^\top \mathbf{x} + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1). \quad (1)$$

The noiseless teacher signal variance is

$$S = (\boldsymbol{\beta}^\star)^\top \Sigma \boldsymbol{\beta}^\star. \quad (2)$$

We first map the input to p linear features,

$$\mathbf{z} = A\mathbf{x} \in \mathbb{R}^p, \quad (3)$$

and fit ridge regression in feature coordinates. With the training feature matrix $Z \in \mathbb{R}^{n \times p}$ and $\widehat{C} = Z^\top Z/n$, the normalized-loss convention is

$$\widehat{\boldsymbol{\theta}}_\lambda = \arg \min_{\boldsymbol{\theta}} \left\{ \frac{1}{n} \|\mathbf{y} - Z\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|^2 \right\} = (\widehat{C} + \lambda I)^{-1} \frac{Z^\top \mathbf{y}}{n}. \quad (4)$$

The resulting model as a function of pixels is

$$\widehat{f}(\mathbf{x}) = \widehat{\boldsymbol{\theta}}^\top A\mathbf{x} = \widehat{\boldsymbol{\beta}}^\top \mathbf{x}, \quad \widehat{\boldsymbol{\beta}} = A^\top \widehat{\boldsymbol{\theta}}. \quad (5)$$

Thus feature-space estimation and pixel-space attribution are not the same object.

1.1 Best population predictor in the feature span

Define

$$C = A\Sigma A^\top, \quad \mathbf{h} = A\Sigma\boldsymbol{\beta}^*. \quad (6)$$

The best linear feature coefficient and its unrepresented residual are

$$\boldsymbol{\theta}^* = C^\dagger \mathbf{h}, \quad (7)$$

$$r(\mathbf{x}) = (\boldsymbol{\beta}^*)^\top \mathbf{x} - (\boldsymbol{\theta}^*)^\top A\mathbf{x}. \quad (8)$$

By the normal equations, $\text{Cov}(\mathbf{z}, r) = 0$. Since the joint distribution is Gaussian, the residual is independent of the retained features. Its variance is

$$S_\perp = \text{Var}(r) = S - (\boldsymbol{\theta}^*)^\top C \boldsymbol{\theta}^*. \quad (9)$$

Consequently, truncated features reduce the training model to

$$y = (\boldsymbol{\theta}^*)^\top \mathbf{z} + \xi, \quad \text{Var}(\xi) = \sigma_{\text{eff}}^2 = \sigma^2 + S_\perp. \quad (10)$$

The omitted teacher signal is simultaneously irreducible generalization error and additional independent training noise.

2 Two geometries: prediction versus pixel control

Natural-image prediction is governed by $C = A\Sigma A^\top$. Pixel-space gradient motion is governed by the distinct matrix

$$G = AA^\top, \quad \mathbf{q} = A\boldsymbol{\beta}^*. \quad (11)$$

The vector $\mathbf{h} = A\Sigma\boldsymbol{\beta}^*$ determines population prediction, whereas $\mathbf{q} = A\boldsymbol{\beta}^*$ determines the teacher's response to a back-propagated pixel gradient. Confusing these two vectors hides the prediction–control dissociation.

2.1 Natural-distribution metrics

For a fixed fitted model, the noiseless population prediction error is

$$E_{\text{gen}}(\hat{\boldsymbol{\theta}}) = S_\perp + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)^\top C (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*). \quad (12)$$

We use the noiseless-teacher coefficient of determination

$$R_{\text{gen}}^2 = 1 - \frac{E_{\text{gen}}}{S}. \quad (13)$$

The slope of true response on predicted response over natural inputs is

$$b_{\text{gen}} = \frac{\text{Cov}((\boldsymbol{\beta}^*)^\top \mathbf{x}, \hat{\boldsymbol{\theta}}^\top A\mathbf{x})}{\text{Var}(\hat{\boldsymbol{\theta}}^\top A\mathbf{x})} = \frac{\mathbf{h}^\top \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^\top C \hat{\boldsymbol{\theta}}}. \quad (14)$$

2.2 Own-path accentuation metrics

Gradient accentuation from a seed $\mathbf{x}_0$ follows

$$\mathbf{x}(t) = \mathbf{x}_0 + t \nabla_{\mathbf{x}} \hat{f}(\mathbf{x}) = \mathbf{x}_0 + t A^\top \hat{\boldsymbol{\theta}}. \quad (15)$$

The student and teacher changes along this path are, respectively,

$$\Delta \hat{f}(t) = t \hat{\boldsymbol{\theta}}^\top G \hat{\boldsymbol{\theta}}, \quad (16)$$

$$\Delta f^\star(t) = t \mathbf{q}^\top \hat{\boldsymbol{\theta}}. \quad (17)$$

Therefore the exact true-on-predicted slope along the path is

$$b_{\text{acc}}(\hat{\boldsymbol{\theta}}) = \frac{\mathbf{q}^\top \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^\top G \hat{\boldsymbol{\theta}}}. \quad (18)$$

Suppose the experimenter chooses the distribution of t so that the student's pathwise response variance equals the natural teacher variance S :

$$\text{Var}(t) = \frac{S}{(\hat{\boldsymbol{\theta}}^\top G \hat{\boldsymbol{\theta}})^2}. \quad (19)$$

The accentuation calibration error is then

$$\frac{E_{\text{acc}}}{S} = (1 - b_{\text{acc}})^2 = \left(1 - \frac{\mathbf{q}^\top \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^\top G \hat{\boldsymbol{\theta}}}\right)^2. \quad (20)$$

When the denominator of R^2 is the variance of the *true* response on the generated path, the exact pathwise statistic becomes

$$R_{\text{acc}}^2 = 1 - \left(1 - \frac{\hat{\boldsymbol{\theta}}^\top G \hat{\boldsymbol{\theta}}}{\mathbf{q}^\top \hat{\boldsymbol{\theta}}}\right)^2. \quad (21)$$

Unlike E_{acc}/S , this statistic is singular when the true path gain $\mathbf{q}^\top \hat{\boldsymbol{\theta}}$ is close to zero. Hence it can be extremely negative even when other summary metrics appear bounded.

3 Spectral feature maps and deterministic equivalents

We now specialize to the setting implemented in the accompanying numerical experiments. Write the input eigendecomposition as

$$\Sigma = U \text{diag}(s_1, \dots, s_d) U^\top, \quad b_j = \mathbf{u}_j^\top \boldsymbol{\beta}^\star. \quad (22)$$

For an index set $\mathcal{J}$, let the feature map retain and rescale PCs,

$$z_j = a_j \mathbf{u}_j^\top \mathbf{x}, \quad j \in \mathcal{J}. \quad (23)$$

In the common eigenbasis of prediction and control, define

$$c_j = a_j^2 s_j, \quad g_j = a_j^2, \quad \theta_j^\star = \frac{b_j}{a_j}. \quad (24)$$

Then $C = \text{diag}(c_j)$, $G = \text{diag}(g_j)$, and $q_j = g_j \theta_j^*$. The omitted signal and effective noise are

$$S_{\perp} = \sum_{j \notin \mathcal{J}} s_j b_j^2, \text{quad} \sigma_{\text{eff}}^2 = \sigma^2 + S_{\perp}. \quad (25)$$

We take $p, n \rightarrow \infty$ with $p/n \rightarrow \gamma$. The effective ridge parameter $\kappa = \kappa(\lambda)$ solves the Silverstein fixed-point equation

$$\kappa - \lambda = \frac{\kappa}{n} \sum_{j \in \mathcal{J}} \frac{c_j}{c_j + \kappa}. \quad (26)$$

Equivalently, the right side is $\gamma \kappa \int c/(c + \kappa) d\nu(c)$ for the empirical feature spectrum ν .

Define

$$h_j = \frac{c_j}{c_j + \kappa}, \quad (27)$$

$$\text{df}_2(\kappa) = \sum_j \frac{c_j^2}{(c_j + \kappa)^2}, \quad (28)$$

$$C_{\text{sig}}(\kappa) = \sum_j \frac{c_j (\theta_j^*)^2}{(c_j + \kappa)^2}, \quad (29)$$

$$v_0(\kappa) = \frac{\kappa^2 C_{\text{sig}}(\kappa) + \sigma_{\text{eff}}^2}{n - \text{df}_2(\kappa)}. \quad (30)$$

The leading componentwise first and second moments of the fitted feature weight are

$$m_j \equiv \mathbb{E}[\hat{\theta}_j] \simeq h_j \theta_j^*, \quad (31)$$

$$v_j \equiv \text{Var}(\hat{\theta}_j) \simeq v_0 \frac{c_j}{(c_j + \kappa)^2}, \quad (32)$$

$$\mathbb{E}[\hat{\theta}_j^2] \simeq m_j^2 + v_j. \quad (33)$$

Thus the per-feature coefficient error is

$$e_j \equiv \mathbb{E}[(\hat{\theta}_j - \theta_j^*)^2] \simeq (1 - h_j)^2 (\theta_j^*)^2 + v_j. \quad (34)$$

The first term is ridge shrinkage bias; the second combines finite-sample signal and response-noise fluctuations.

3.1 DE generalization metrics

Aggregating the componentwise moments gives

$$E_{\text{gen}}^{\text{DE}} = S_{\perp} + \sum_j c_j e_j, \quad (35)$$

$$R_{\text{gen}}^{2, \text{DE}} = 1 - \frac{E_{\text{gen}}^{\text{DE}}}{S}, \quad (36)$$

$$N_{\text{gen}}^{\text{DE}} = \sum_j c_j m_j \theta_j^*, \quad (37)$$

$$D_{\text{gen}}^{\text{DE}} = \sum_j c_j (m_j^2 + v_j), \quad (38)$$

$$b_{\text{gen}}^{\text{DE}} = \frac{N_{\text{gen}}^{\text{DE}}}{D_{\text{gen}}^{\text{DE}}}. \quad (39)$$

3.2 DE pixel-accentuation metrics

The control geometry changes only the aggregation weights:

$$N_{\text{acc}}^{\text{DE}} = \sum_j g_j m_j \theta_j^*, \quad (40)$$

$$D_{\text{acc}}^{\text{DE}} = \sum_j g_j (m_j^2 + v_j), \quad (41)$$

$$b_{\text{acc}}^{\text{DE}} = \frac{N_{\text{acc}}^{\text{DE}}}{D_{\text{acc}}^{\text{DE}}}, \quad (42)$$

$$\frac{E_{\text{acc}}^{\text{DE}}}{S} = (1 - b_{\text{acc}}^{\text{DE}})^2, \quad (43)$$

$$R_{\text{acc}}^{2,\text{DE}} = 1 - \left(1 - \frac{D_{\text{acc}}^{\text{DE}}}{N_{\text{acc}}^{\text{DE}}}\right)^2. \quad (44)$$

Equations (35)–(44) are the main feature-space extension. Natural generalization weights mode j by c_j ; pixel accentuation weights it by g_j . The two agree only in special cases.

The DE above substitutes ratios of deterministic first and second moments for the nonlinear per-fit ratios. It predicts a typical high-dimensional fit but is not, in general, the exact Monte Carlo mean of R_{acc}^2 . This distinction matters whenever N_{acc} can approach zero: rare fits then dominate the empirical mean, while the median can remain close to the leading DE.

3.3 Input-space weight recovery

The recovered pixel coefficient in retained mode j is $\hat{b}_j = a_j \hat{\theta}_j$. Therefore

$$\mathbb{E}[\hat{b}_j] \simeq a_j m_j, \quad (45)$$

$$\text{Var}(\hat{b}_j) \simeq g_j v_j, \quad (46)$$

$$\mathbb{E}\|\hat{\beta} - \beta^*\|^2 \simeq \sum_{j \in \mathcal{J}} g_j e_j + \sum_{j \notin \mathcal{J}} b_j^2. \quad (47)$$

This formula makes the high-frequency mechanism explicit. A low-variance direction can make a negligible contribution $s_j e_j$ to prediction error but a large contribution $g_j e_j$ to the back-propagated pixel gradient. The DE predicts the mean and variance envelope across modes; it does not predict the signs or texture of one particular random fitted realization.

4 Cross-validated ridge selection from the DE

Many software packages parameterize ridge as

$$\|\mathbf{y} - Z\boldsymbol{\theta}\|^2 + \alpha \|\boldsymbol{\theta}\|^2, \quad (48)$$

so that $\lambda = \alpha/n$ in Eq. (4). For analytic leave-one-out cross-validation, a DE approximation follows the actual training-size change:

1. set $n_{\text{tr}} = n - 1$;
2. for every candidate α , solve Eq. (26) with $\lambda = \alpha/n_{\text{tr}}$ and aspect ratio p/n_{tr} ;

3. compute $E_{\text{gen}}^{\text{DE}}(\alpha; n_{\text{tr}})$ using σ_{eff}^2 ;
4. select $\alpha_{\text{DE-CV}} = \arg \min_{\alpha} E_{\text{gen}}^{\text{DE}}$;
5. refit the final n -sample DE with $\lambda = \alpha_{\text{DE-CV}}/n$.

For K -fold CV, replace $n - 1$ by the fold training size. A dense log-spaced α grid is important: a coarse grid creates artificial jumps in the selected regularization and therefore jagged downstream control metrics. The DE selects ridge for natural predictive risk, just as ordinary CV does; it does not secretly optimize accentuation.

5 PCA, whitening, and top-PC truncation

The spectral formulas expose the behavior of the three feature spaces used in the FFHQ disk-teacher experiment.

Feature map	a_j	c_j	g_j	omitted signal
Full PCA	1	s_j	1	0
Full whitening	$s_j^{-1/2}$	1	s_j^{-1}	0
Top K PCs	$1, j \leq K$	s_j	1	$\sum_{j>K} s_j b_j^2$

Table 1: Prediction and pixel-control quantities for aligned linear features.

Full PCA. An orthogonal change of basis leaves ridge and the pixel-gradient norm unchanged. It is the pixel-space baseline written in interpretable spectral coordinates.

Full whitening. Regression sees an isotropic covariance, $c_j = 1$, but back-propagation weights each mode by $g_j = 1/s_j$. Low-variance, high-frequency PCs can therefore dominate D_{acc} even though they do not dominate natural prediction. The feature representation may look well conditioned while pixel control is ill conditioned.

Top-PC truncation. Discarding low-variance PCs introduces irreducible natural error $S_{\perp}$, but completely prevents those directions from entering the pixel-gradient path. When the teacher signal is concentrated in the leading PCs, a modest predictive cost can buy a large control benefit. The optimal cutoff depends on noise: higher response noise generally favors a smaller, more signal-focused feature space.

6 Peer review across models and feature spaces

Let an evaluator have pixel weight $\hat{\mathbf{w}}_e = A_e^{\top} \hat{\boldsymbol{\theta}}_e$ and a generator have $\hat{\mathbf{w}}_g = A_g^{\top} \hat{\boldsymbol{\theta}}_g$. Along the generator’s path $\mathbf{x}(t) = \mathbf{x}_0 + t\hat{\mathbf{w}}_g$, the teacher and evaluator gains are

$$T_g = (\boldsymbol{\beta}^*)^{\top} \hat{\mathbf{w}}_g, \quad D_{e \leftarrow g} = \hat{\mathbf{w}}_e^{\top} \hat{\mathbf{w}}_g = \hat{\boldsymbol{\theta}}_e^{\top} A_e A_g^{\top} \hat{\boldsymbol{\theta}}_g. \quad (49)$$

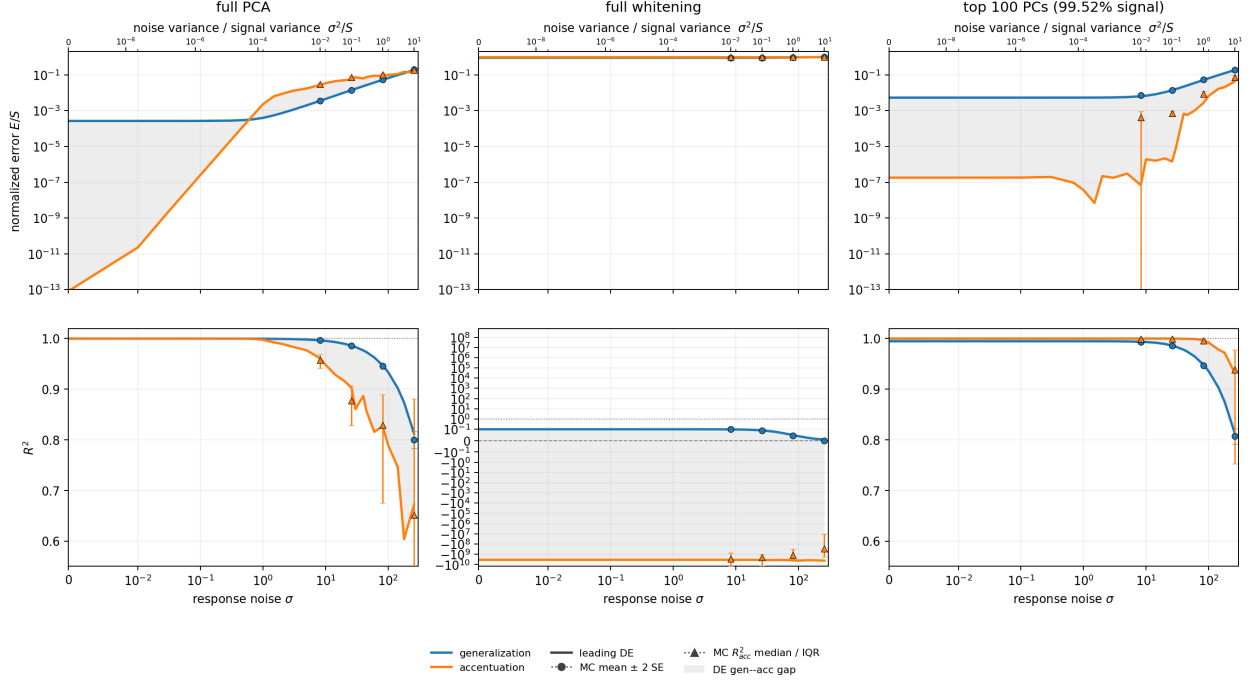


Figure 1: Measured-spectrum FFHQ disk-teacher experiment. Curves are the leading DE; points are Monte Carlo ridge fits. Full PCA develops a generalization–accentuation gap, top-100-PC truncation suppresses the harmful control directions, and full whitening produces a near-zero true path gain and extremely negative R^2_{acc} . The lower horizontal axis is response-noise standard deviation; the upper axis is σ^2/S .

If the generator scales its path to its own natural-response target range, the evaluator’s normalized peer error, slope, and R^2 are

$$\frac{E_{e \leftarrow g}}{S} = \left(\frac{D_{e \leftarrow g} - T_g}{\widehat{\mathbf{w}}_g^\top \widehat{\mathbf{w}}_g} \right)^2, \quad (50)$$

$$b_{e \leftarrow g} = \frac{T_g}{D_{e \leftarrow g}}, \quad (51)$$

$$R^2_{e \leftarrow g} = 1 - \left(1 - \frac{D_{e \leftarrow g}}{T_g} \right)^2. \quad (52)$$

For two independently trained models, a leading peer DE can replace each weight by its deterministic mean in $D_{e \leftarrow g}$ and T_g . Models trained on the same or overlapping samples require a joint two-resolvent DE to retain their correlated fluctuations. Equation (52) also clarifies the observed asymmetry: a generator can move in directions that remain legible to a good evaluator, while the reverse path can have nearly zero teacher gain.

7 Interpretation and limitations

The main conclusions are:

1. Good natural-image prediction does not imply good pixel-space control. Prediction uses C ; control uses G .
2. Low-variance modes are cheap under natural risk but can be expensive under pixel-gradient accentuation, especially after whitening.
3. Top-PC truncation can reverse the generalization–accentuation gap by removing directions that contribute little teacher signal but large control variance.
4. The feature-space DE predicts mean per-mode bias and variance, aggregate natural risk, typical control calibration, and DE-CV ridge selection. It does not specify the visual pattern of a single random fit.
5. R_{acc}^2 and peer R^2 are nonlinear ratios. Near-zero true path gain creates heavy tails, so empirical medians/IQRs and means/standard errors should both be reported.

The present closed-form feature DE assumes a linear feature map aligned with the input covariance eigenbasis, so that C and G commute. The exact finite-sample formulas in Sections 1–2 hold for arbitrary linear A . Extending the DE to noncommuting C and G , nonlinear learned features, and joint peer-review fluctuations requires operator-valued or bilinear-resolvent deterministic equivalents.